Learning Humanoid Recovery Under Overhead Constraints

Anonymous authors

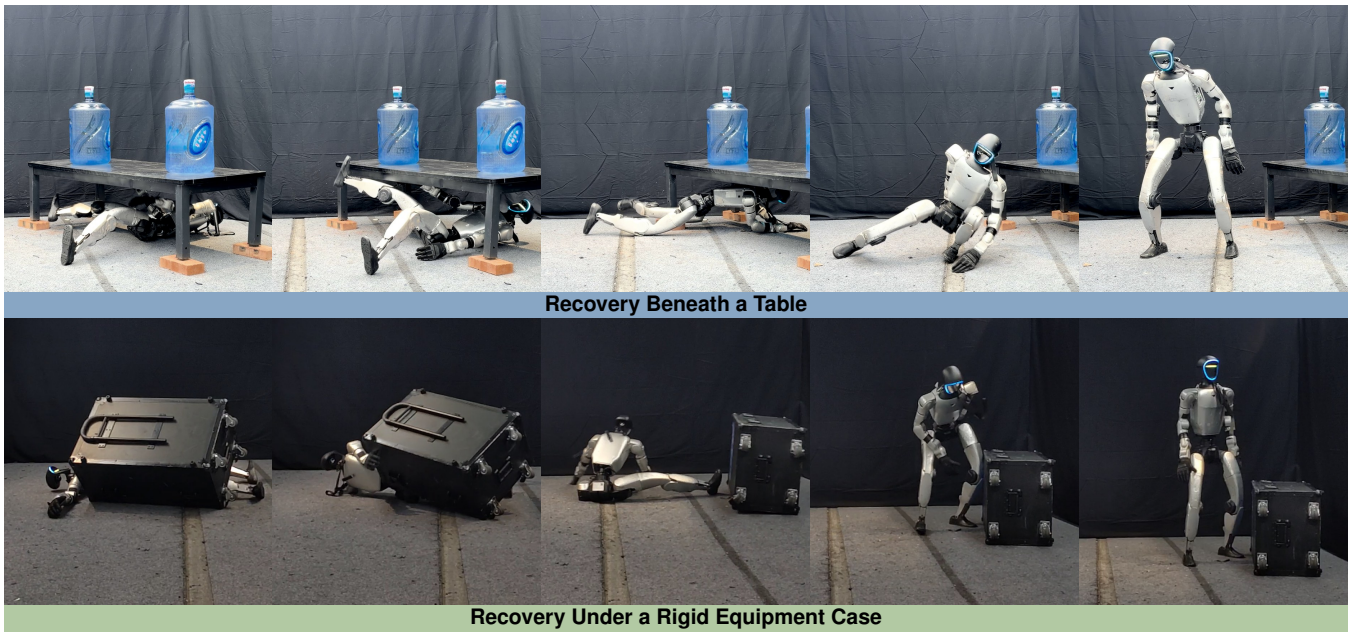


Fig. 1. Recovery under overhead constraints. Beneath a table (top), the robot adjusts its support and clears the table edge before standing. Beneath a rigid equipment case (bottom), interaction tips the obstacle aside and permits the rise. Each row shows successive frames from one continuous hardware trial.

***Abstract*— A fallen humanoid beneath an overhead obstacle faces a coupled recovery problem. It must create clearance while organizing support for a stable rise, since attempting to stand before making room can further increase obstruction. We introduce a reinforcement learning method with contact-supported geometric escape guidance, which rewards incremental reduction of robot-obstacle overlap without prescribing an escape path. To achieve this, a learned motion prior, together with posture-dependent support and joint-stall costs, is proposed to shape a single policy that learns when to clear the obstacle, how to establish support, and when to complete the rise, all using proprioception alone. In simulation, it recovers in 88.5% of trials beneath plates that can only be lifted, compared with 18.9% without geometric guidance, and in 98.6% of trials beneath free plates with randomized size, mass and placement. On a Unitree G1, the deployed policy recovered in 95 of 150 attempts across seven real-world obstacle scenes, including furniture and multi-object arrangements absent from training.**

I. Introduction

Humanoid robots that operate in homes, offices, and disaster sites will occasionally fall, and in cluttered spaces, they may come to rest beneath a table, a shelf, or a fallen board. A standard get-up motion often fails in this situation: raising the torso pushes the body into the obstacle above it. The robot may need to move laterally or displace the obstacle to create sufficient room to rise. Fig. 1 shows two such recoveries on hardware.

Learning-based recovery has advanced rapidly. Recent humanoid policies get up from diverse fallen postures and on varied terrain, and some combine recovery with fall mitigation or locomotion [1], [2], [3], [4], [5]. These methods, however, assume open space above the robot. Under an overhead obstacle, a reward for raising the head and torso works against the low, sideways movement needed to escape.

We address this problem by rewarding the robot for making room before it stands. During training, a geometric reward measures how much of the body still lies beneath the obstacle, how large a gap has opened, and how far the root has moved from the obstacle. Its main term rewards each new reduction in coverage and each new widening of the gap, but only while the robot stays low and supports itself with its hands. It does not prescribe an escape direction or path. Rewards for standing stay active throughout, so the robot retains an incentive to rise as sufficient space becomes available. A motion prior learned from human get-up data encourages natural, coordinated movement, and two additional costs discourage foot slipping near the end of the rise and joints that stay heavily loaded without moving. The result is a single policy that uses only proprioception; it needs no obstacle map, planned trajectory or switching between controllers.

Training progresses from flat-ground recovery to obstacle interaction, with initial states sampled from human get-up

motions and procedural fallen poses. In simulation, the policy stands in 88.5% of trials beneath a plate that can only be lifted, compared with 18.9% without the geometric reward. On a Unitree G1, it recovers in 95 of 150 attempts beneath seven real-world obstacles, including furniture and multi-object arrangements absent from training.

Our contributions are as follows:

- We study humanoid recovery beneath overhead obstacles, where the robot must create sufficient space to regain a stable standing posture, and develop a single policy that uses only proprioception at deployment.
- We propose a geometric reward that measures coverage, clearance and root separation without prescribing an escape path. Its main term pays only for progress made while the robot is low and supported by its hands, and additional costs improve foot support and joint loading during the rise.
- We compare against existing humanoid recovery methods and ablate the geometric reward and two execution costs in simulation, and we evaluate the deployed policy in 170 hardware trials, 150 of them beneath seven obstacle arrangements.

II. Related Work

A. Learning to Recover

Humanoid stand-up has progressed from learning a feasible get-up strategy to making recovery robust and deployable. Early work by Morimoto and Doya demonstrated hierarchical reinforcement learning for stand-up behavior on a real robot [6]. later works learned quadruped recovery [7] and integrated rapid recovery with locomotion and task behavior on a small bipedal soccer robot [8]. More recent humanoid work addresses the demands of larger platforms. HumanUP discovers get-up motions and then refines them for safe deployment across postures and terrains [1], while HoST learns posture-adaptive recovery with a curriculum and motion constraints [2]. FIRM extends the scope from standing up after a fall to a unified fall-safety policy, using demonstrations and diffusion-based memory to connect fall mitigation with recovery [3].

These methods show that robots can recover across varied postures, terrain and fall conditions, yet, they do not address recovery when an overhead obstacle occupies the space needed to rise, where the robot must first create clearance through contact while preserving the support needed to stand.

B. Contact as Support and Obstruction

Environmental contact can help robots recover, but it can also constrain the motion needed to stand. Contact-planning methods explicitly reason about how contact transitions redirect motion or change the support configuration. [9] proposes to plan contacts to dissipate falling momentum, while contact-transition trees jointly optimize stabilization trajectories and support changes [10]. [11] incorporates sliding and detaching contacts into motion planning, including for stand-up motions. Learning-based methods likewise use non-foot contacts, where the arm contact can reduce fall damage and assist recovery in a legged manipulator [12].

The learning-based methods extend recovery to varied terrain and contact opportunities. VIGOR distills terrain-aware recovery goals into a depth-conditioned policy [4], while UniReLo uses terrain-relative support guidance and motion priors to support recovery across heterogeneous terrain [5]. RecoverFormer predicts useful environmental contact regions for balance recovery [13]. Whole-body planning for confined spaces addresses geometric constraints during traversal [14]. Nevertheless, recovery beneath an overhead obstacle poses an unique challenge, where the obstacle both blocks the motion needed to rise and provides a surface the robot can push or lift to create clearance.

C. Motion Priors and Reusable Skills

Motion priors address a different part of recovery, where they encourage coordinated and human-like movement while task rewards specify what the robot must accomplish. DeepMimic combines reference-motion tracking with task objectives [15]. Adversarial methods such as GAIL and AMP learn motion rewards from examples rather than requiring direct trajectory tracking [16], [17], [18]. ASE and CALM extend this idea by learning skill representations that can be reused across downstream tasks [19], [20]. More recent work applies motion priors to a wide range of humanoid behaviors. The State-dependent AMP combines locomotion and recovery [21], while StableMimic and Heracles connect motion tracking with recovery generation [22], [23].

Diffusion models can serve as action policies [24] or as frozen motion-prior reward models, as in SMP [25]. We use SMP to encourage coordinated movement during training, and remove it at deployment. Because a general motion prior does not specify how to clear a particular obstacle, we pair it with geometric escape guidance that rewards obstacle-specific clearance progress.

D. Execution Quality and Physical Transfer

Successful motion in simulation can still be oscillatory or sensitive to dynamics. Dynamics randomization addresses model mismatch [26], while Lipschitz-constrained policies directly regularize the sensitivity of humanoid actions [27]. Recovery methods also constrain motion speed and refine discovered trajectories [2], [1]. Our foot-motion and joint-stall costs address two problems that arise late in recovery, where feet keep slipping as the robot becomes upright and joints stay heavily loaded without making progress.

III. Problem Formulation

We consider a humanoid lying on flat ground, possibly beneath an overhead obstacle. The policy is a mapping $\pi_\theta : o_t \mapsto a_t$ from proprioceptive observations $o_t \in \mathbb{R}^{93}$ to joint-position targets $a_t \in \mathbb{R}^{29}$. During training, the obstacle geometry and contacts enter the reward r_t, but the deployed policy does not observe them, where o_t contains no obstacle state. We train with two plate types that differ in

their admissible motions $\mathcal{M}_{\text{obs}}$, and hence in what escape requires. A *guided plate* admits only vertical translation, $\mathcal{M}_{\text{obs}} = \{(0, 0, \delta z)\}$, so the robot can lift it but cannot slide or tilt it, and lateral repositioning is the only way to create room to rise. A *free plate* is unattached, $\mathcal{M}_{\text{obs}} = \text{SE}(3)$, so the robot can push, slide or tip it aside, coupling escape to contact-rich manipulation. Training also includes episodes without obstacles. Fixed obstacles such as tables and multi-object arrangements are never seen during training where we use them to evaluation generalization on hardware.

Our goal is to stand and remain standing, where the complete separation from the obstacle is not required. Escaping the obstacle and standing are related but distinct. The robot might push a plate aside and still fail to rise, or stand while still touching the plate. Success is thus judged only by standing, and escape enters only the training reward as a shaping signal.

IV. Method

Recovery beneath an overhead obstacle requires the robot to make room to move while still learning to stand. We train one policy to handle both. A curriculum first teaches flat-ground recovery and controlled execution, then introduces obstacle interaction. During obstacle training, we propose geometric rewards to provide feedback about clearance; at deployment, the actor receives proprioception alone. A learned motion prior encourages coordinated movement, while posture-dependent costs let the robot move freely when it is low, and favor controlled support as it rises.

A. Policy and Training Curriculum

Our deployed policy is the proprioceptive actor coupled to a PD controller. At each control step, the actor converts the current observation into joint-position targets. The actor maps proprioceptive observations to joint-position targets. It observes base angular velocity ω_t^b, projected gravity g_t^b, joint-position offsets $(q_t - q^0)$, joint velocities $\dot{q}_t$ and its previous action a_{t-1}:

$$o_t = [\omega_t^b, g_t^b, q_t - q^0, \dot{q}_t, a_{t-1}] \in \mathbb{R}^{93}. \tag{1}$$

The actor outputs 29 joint-position targets at 50Hz. The PD controller track these targets in MuJoCo [28], which integrates the robot dynamics at 500Hz.

Recovery must be learned across the full transition from lying down to standing. We thus initialize episodes from retargeted LAFAN1 get-up clips [29], grouped by head height and uprightness into low ($\mathcal{L}$), intermediate ($\mathcal{M}$) and near-standing ($\mathcal{H}$) states. These groups determine reset states, where the policy does not track the clips. Within the $\mathcal{L}$ group, 75% of resets come from the clips and 25% from perturbed procedural poses representing supine, prone, left-side and right-side falls. $\mathcal{M}$ and $\mathcal{H}$ resets come from the clips. We stratify sampling by posture to compensate for uneven coverage in the source data, reject invalid configurations and reset the robot with zero velocity.

TABLE I

Initial-state distribution across training stages.

Stage	Scene share	$\mathcal{L}$	$\mathcal{M}$	$\mathcal{H}$
Stage I (Basic)	Flat: 100%	20	40	40
Stage II (Refine)	Flat: 100%	40	40	20
Stage III (Adapt)	Flat: 50%	40	40	20
	Guided plate: 25%	100	0	0
	Free plate: 25%	100	0	0

The policy is trained with PPO [30] using an asymmetric actor-critic. The critic uses robot-state observations, including base linear velocity, obstacle state and contact labels, to estimate the expected cumulative discounted reward from the current state. These quantities are unavailable to the actor. At deployment, only the actor and PD controller are used. To be specific, training contains stages:

- **Stage I** learns basic upward recovery on flat ground from a randomly initialized actor, using rewards for upward motion and height.
- **Stage II** refines the motion by introducing posture-dependent recovery and quiet-standing objectives together with execution costs; it also increases the proportion of low starts.
- **Stage III** adapts the policy to obstacle interaction while retaining flat-ground recovery, where half of the environments remain on flat ground, one quarter contain a guided plate, and one quarter contain a free plate. All obstacle episodes begin from low ($\mathcal{L}$) states.

The four fallen-posture families are equally likely in Stages I and III. Stage II emphasizes prone starts, sampled with a probability of 50%, while the other three families share the remaining probability equally. Reset groups and within-episode reward phases serve different purposes, where $\mathcal{L}$, $\mathcal{M}$ and $\mathcal{H}$ control where an episode begins, while the phases described below shape behavior as the robot recovers.

B. Reward Structure

We introduce the reward components progressively as the policy learns to recover. Stage I uses simple rewards for upward motion and height. In Stage II, we use a fixed SMP motion prior [25] together with posture-dependent objectives for rising and settling. The prior reward S_t is computed from normalized diffusion prediction errors, where higher values indicate motion more consistent with the human get-up data. Its normalization scales are estimated in Stage I and fixed thereafter. Multiplying the task objectives by S_t favors recovery that is both successful and consistent with the motion prior.

The task objectives contain a rising term T_t^{rise} and a quiet-motion term T_t^{quiet}, as detailed in Table II. The rising term rewards movement toward an upright posture, while the quiet-motion term encourages the robot to settle once it establishes support. Stage II also introduces execution costs. Stage III retains these objectives and costs, adds the foot-motion cost Q_t and joint-stall cost L_t, and introduces the

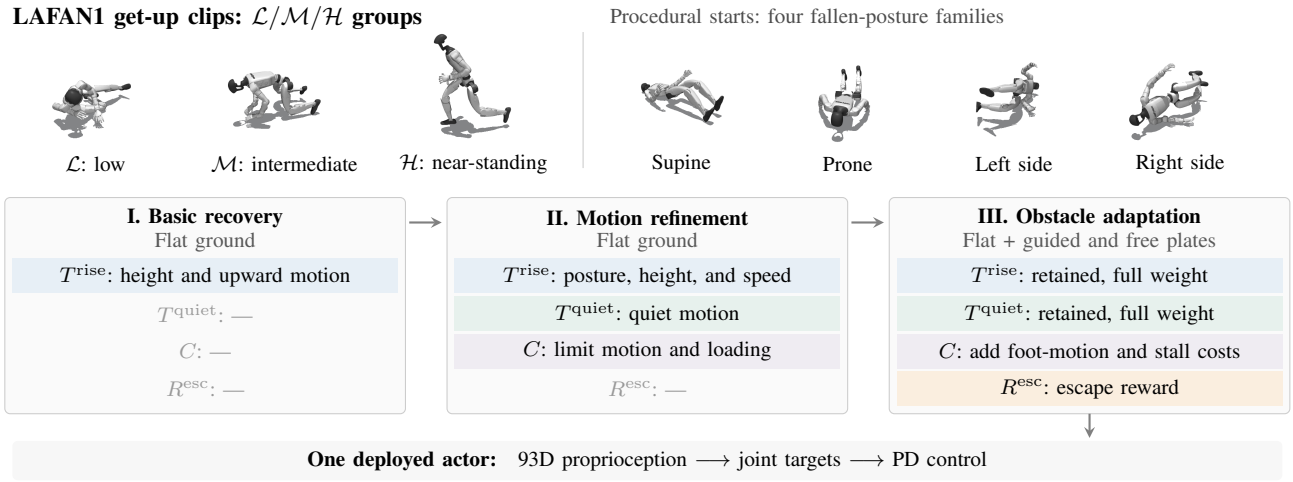


Fig. 2. Training overview. Top: example initial states from each $\mathcal{L}/\mathcal{M}/\mathcal{H}$ group and fallen-posture family. Bottom: reward groups active in each training stage, in the order and colors of Table II; dashes mark absent groups. The actor and its observation normalization carry over between stages, and a single actor is deployed.

obstacle-escape reward R_t^{esc}. Thus, the final-stage reward is

$$r_t = S_t\big(T_t^{\text{rise}} + T_t^{\text{quiet}}\big) - C_t + R_t^{\text{esc}}, \tag{2}$$

where $C_t \geq 0$ collects all execution costs.

The foot-motion cost Q_t and joint-stall cost L_t target two specific execution problems. The quiet-feet reward $r_{\text{quiet feet}}$ already favors low foot speed, but its effect is bounded and scaled by the motion-prior reward S_t. The direct cost Q_t thus penalizes foot motion near upright postures. The cost L_t penalizes joints that remain highly loaded without moving, as can happen when the robot pushes against a plate that does not yield. Thus, the motion prior encourages coordinated movement, while Q_t and L_t directly penalize foot slipping and stalled joint loading.

Within each recovery episode, s denotes the current reward phase: low, transition, rising, or standing. Unlike training stages I–III, these phases change with the robot's posture during a single recovery. Each phase specifies a target head height $h_{\text{head},s}^{\text{target}}$, target torso uprightness $u_{\text{torso},s}^{\text{target}}$, an allowed knee range, and an upward-speed target. The initial phase is determined by the reset posture. Progression through transition and rising requires height, uprightness, knee-angle, and vertical-speed conditions to hold for 0.2 s; entering standing additionally requires support from both feet and a 0.5 s confirmation. A fall returns the episode to the low phase. As the robot rises, action and torque costs increase, and joint-speed and power limits tighten. Foot- and base-motion penalties increase near upright postures, preserving freedom to move near the floor. These reward phases shape the reward but are not provided to the actor.

The upward-speed target tapers as the robot approaches its target height, allowing the rise to settle rather than stop abruptly. Motion and loading costs are relaxed near the floor, where larger movements may be needed to establish support. Table II gives the reward definitions and stage-specific weights.

C. Geometric Guidance for Obstacle Clearance

Standing rewards alone provide limited guidance about how to escape from beneath an obstacle. We thus add an escape objective R_t^{esc} during Stage III. It combines three geometric signals, including supported progress, dense clearance and root-plate separation, with an escape-completion reward and an excess-force penalty. The geometric signals describe different aspects of progress, which includes reducing body-plate overlap, maintaining clearance, and moving the robot away from the plate. Only the supported-progress term explicitly gates progress on low posture and hand support.

To compute supported progress, we project the bounding boxes of the robot's 29 collision shapes and the plate onto the ground plane, aligned with the world axes, as shown in Figure 3. For each projected robot box i that overlaps the plate box, let o_{ix} and o_{iy} be the distances it would need to move along the world x- and y-axes to leave the plate box. We define its covered depth as

$$d_i = \min(o_{ix}, o_{iy}), \tag{3}$$

and set $d_i = 0$ when the boxes do not overlap. The total covered depth is $D_t = \sum_i d_i$, which reaches zero when no projected body box overlaps the projected plate box.

We also measure the clearance gap c_t, defined as the smallest horizontal distance between any projected body box and the plate box. It is zero while any box overlaps the plate and becomes positive after all projected boxes separate. Thus, D_t provides a progress signal while the robot is still partly beneath the plate, whereas c_t measures separation after overlap ends. Using individual collision shapes allows the reward to register a limb clearing the plate before the whole body has moved out. These quantities are planar approximations: axis-aligned bounds can overestimate overlap and underestimate gaps for rotated shapes, and they do not test vertical ordering. In particular, the projected measures include body shapes that are entirely above the plate; the escape criterion handles that case separately.

To reward new progress without repeatedly paying for the same configuration, we compare the current measurements

TABLE II

REWARD DEFINITIONS AND SIGNED WEIGHTS ACROSS THE THREE TRAINING STAGES. COLORS FOLLOW FIG. 2; DASHES INDICATE INACTIVE TERMS. WEIGHTS ARE SCALED BY THE CONTROL TIMESTEP (0.02 S).

Term	Definition	I: Basic	II: Refine	III: Adapt
T^{rise}: recovery				
Basic upward motion	$\begin{cases} e^{-100[0.25-\dot{h}_{\mathrm{head}}]_+^2}, & h_{\mathrm{head}} < 0.9 \\ 1, & h_{\mathrm{head}} \geq 0.9 \end{cases}$	0.70	–	–
Basic height	$e^{-[1.1-h_{\mathrm{head}}]_+^2}$	0.30	–	–
Phase-dependent posture	$e^{-8[h_{\mathrm{head},s}^{\mathrm{target}}-h_{\mathrm{head}}]_+^2-6[u_{\mathrm{torso},s}^{\mathrm{target}}-u_{\mathrm{torso}}]_+^2-5e_{\mathrm{knee}}}$	–	0.22	0.22
Head velocity	$e^{-45(\dot{h}_{\mathrm{head}}-v_{\mathrm{head},s}^{*})^2}\, e^{-140([\dot{h}_{\mathrm{head}}-v_{\mathrm{head},s}^{\mathrm{up}}]_+^2+[-\dot{h}_{\mathrm{head}}-v_{\mathrm{head},s}^{\mathrm{down}}]_+^2)}$ where $v_{\mathrm{head},s}^{*}=\bar{v}_{\mathrm{head},s}\chi((h_{\mathrm{head},s}^{\mathrm{target}}-h_{\mathrm{head}})/\delta_h)$	–	0.18	0.18
Head height	$e^{-2[1.15-h_{\mathrm{head}}]_+^2}$	–	0.10	0.10
Torso uprightness	$\left(u_{\mathrm{torso}}\right)^2$	–	0.15	0.15
T^{quiet}: quiet motion				
Quiet feet	$r_{\mathrm{quiet\,feet}}=1-g_{\mathrm{upright}}+g_{\mathrm{upright}}e^{-20\overline{v_{\mathrm{foot}}^2}}$ where $g_{\mathrm{upright}}=\sigma(\ell(h_{\mathrm{head}};h_{\mathrm{on}},h_{\mathrm{full}}))\,\sigma(\ell(u_{\mathrm{torso}};u_{\mathrm{on}},u_{\mathrm{full}}))$	–	0.08	0.08
Quiet base	$1-g_{\mathrm{upright}}+g_{\mathrm{upright}}e^{-8\|v_{\mathrm{base}}\|^2}$	–	0.07	0.07
Angular quietness	$e^{-0.8\|\omega_{\mathrm{base}}\|^2}$	–	0.07	0.07
Joint quietness	$e^{-0.04\,\mathrm{mean}_j\,\dot{q}_j^2}$	–	0.06	0.06
Action smoothness	$e^{-4\,\mathrm{mean}_j(\Delta a_j)^2}$	–	0.07	0.07
C: execution costs				
Action rate†	$k_s^{\Delta a}\sum_j(\Delta a_j)^2$	–	-0.0015	-0.0015
Action acceleration†	$k_s^{\Delta^2 a}\sum_j(\Delta^2 a_j)^2$	–	-0.0012	-0.0012
Joint acceleration†	$\langle k_s^{\ddot{q}}\sum_j\ddot{q}_j^2\rangle$	–	-5×10^{-8}	-5×10^{-8}
Torque†	$\langle k_s^{\tau}\sum_j\tau_j^2\rangle$	–	-10^{-6}	-10^{-6}
Joint overspeed†	$\langle\sum_j[\|\dot{q}_j\|-\dot{q}_s^{\max}]_+^2\rangle$	–	-0.02	-0.02
Joint overpower†	$\langle\mathrm{mean}_j[\|\tau_j\dot{q}_j\|-P_s^{\max}]_+^2\rangle$	–	-2×10^{-6}	-2×10^{-6}
Head overspeed†	$\langle[\dot{h}_{\mathrm{head}}-v_{\mathrm{head},s}^{\mathrm{limit}}]_+^2+[-\dot{h}_{\mathrm{head}}-0.20]_+^2\rangle$	–	-1	-1
Sustained effort†	$\left\langle\frac{1}{3}\sum_{j\in\mathcal{J}_{\mathrm{high}}}c_j^{\mathrm{effort}}\right\rangle$ where $c_j^{\mathrm{effort}}=([\sqrt{m_j^{\mathrm{load}}}-0.45]_+/0.55)^2$	–	-0.05	-0.05
Soft joint limits	$\sum_j([q_j^{\min}-q_j]_+ + [q_j-q_j^{\max}]_+)$	–	-0.10	-0.10
Foot motion	$Q_t=\ell(h_{\mathrm{head}};h_{\mathrm{on}},h_{\mathrm{full}})\ell(u_{\mathrm{torso}};u_{\mathrm{on}},u_{\mathrm{full}})\min(\overline{v_{\mathrm{foot}}^2}/0.1^2,10)$	–	–	-0.03
Joint stall	$L_t=\max_j\{\langle c_j^{\mathrm{stall}}\rangle\chi(1-A_j/0.10)\}$ where $c_j^{\mathrm{stall}}=\ell(\rho_j;\rho_{\mathrm{on}},\rho_{\mathrm{full}})^2\ell(T_j;T_{\mathrm{on}},T_{\mathrm{full}})\chi(1-\|\dot{q}_j\|/\dot{q}_{\mathrm{stall}})$	–	–	-0.20
R^{esc}: escape reward				
Supported progress	G_t in (4): best-so-far depth/gap improvement	–	–	0.45
Dense clearance	$J_t\{0.85\chi(1-N_t/\max(N_0,1))+0.15\chi(c_t/0.04)\}$	–	–	0.08
Root–plate separation	$H_t\chi\big((d_{\mathrm{root,plate}}(t)-d_{\mathrm{root,plate}}^{\max}(t))/0.025\big)$	–	–	0.01
Escape completion	$E_t=\mathbf{1}$[escape currently confirmed]	–	–	0.60
Excess plate force	$B_t([F_t-300]_+/300)^2$	–	–	-0.03

$[x]_+=\max(x,0)$, $\chi(x)=\mathrm{clip}(x,0,1)$, $\ell(x;a,b)=\chi((x-a)/(b-a))$, and $\sigma(x)=x^2(3-2x)$. s indexes the within-episode recovery phase (low, transition, rising, standing), not training stages I–III. g_{upright} gates quiet feet/base rewards by head height and torso uprightness. Subscripts on and full denote the onset and full activation of a ramp. $h_{\mathrm{head}}, \dot{h}_{\mathrm{head}}$: head height and vertical speed; $u_{\mathrm{torso}}=\chi(-g_z^b)$: torso uprightness; $\overline{v_{\mathrm{foot}}^2}=\mathrm{mean}_{i\in\mathrm{feet}}\|v_i\|^2$: mean squared foot speed. $h_{\mathrm{head},s}^{\mathrm{target}}, u_{\mathrm{torso},s}^{\mathrm{target}}$: phase posture targets; $h_{\mathrm{on/full}}, u_{\mathrm{on/full}}$: height/uprightness ramp bounds; e_{knee}: knee-range violation; δ_h: speed-taper height margin; $v_{\mathrm{head},s}^{\mathrm{up}}, v_{\mathrm{head},s}^{\mathrm{down}}$: upward/downward head-speed bounds in the task reward; $v_{\mathrm{head},s}^{\mathrm{limit}}$: upward head-speed limit in the execution cost. $\rho_j=|\tau_j|/\tau_j^{\max}$: load ratio; T_j: continuous duration above ρ_{on}; $\dot{q}_{\mathrm{stall}}$: stall-speed threshold; A_j: recent joint excursion; $\langle\cdot\rangle$: physics-substep average. $\Delta a_t=a_t-a_{t-1}$, $\Delta^2 a_t=\Delta a_t-\Delta a_{t-1}$; m_j^{load}: filtered squared load ratio; $\mathcal{J}_{\mathrm{high}}$: indices of the three largest c_j^{effort} at each physics substep; k_s: phase-dependent cost scales. In stage II, costs marked † are ramped in by $\eta=\min(n_{\mathrm{update}}/500,1)$ over the first 500 updates; stage III uses $\eta=1$.

with the best values reached earlier in the episode. Let ΔD_t be the reduction in covered depth below its previous minimum, and let Δc_t be the increase in clearance gap above its previous maximum. The supported-progress reward is

$$G_t = I_t h_t \mathbf{1}[h_{\mathrm{head},t}\leq 0.9]\big[\,\mathrm{clip}(\Delta D_t/0.025,0,1) + \tfrac{1}{2}\,\mathrm{clip}(\Delta c_t/0.02,0,1)\big], \tag{4}$$

where distances are in meters, $h_{\mathrm{head},t}$ is head height, h_t is the fraction of hands providing support, and I_t indicates a valid plate episode in which the robot has contacted the plate but has not yet escaped. The head-height gate limits this reward to low postures, and the hand-support factor favors supported clearance progress. The reward does not specify a crawling direction or escape path.

The remaining terms in R_t^{esc} complement this incremental signal (Table II). Dense clearance rewards the current

fraction of collision shapes no longer beneath the plate and the current gap, so it remains active while clearance is maintained. Here N_t and N_0 are the current and initial counts of shapes beneath the plate. Root–plate separation rewards new episode-wise maxima of the horizontal distance $d_{\mathrm{root,plate}}$ between the robot root and plate center. Escape completion pays while escape remains confirmed: the robot must be clear of the plate and out of contact for 0.3 s; returning beneath the plate removes the reward. Finally, the excess-force term penalizes the largest plate-contact force norm F_t above 300 N. The indicators B_t, J_t, and H_t in Table II select plate episodes, valid plate episodes, and plate episodes with prior contact, respectively.

D. Optimization and Domain Randomization

We train 4096 parallel environments with PPO. Between stages, we transfer the actor weights and observation normalization, and reinitialize the critic and optimizer. In Stages II and III, we disable early termination for reaching a standing state or receiving a low motion-prior score. This allows the policy to continue practicing after a failed attempt and after rising.

For sim-to-real transfer, we randomize observation noise, external pushes, friction, encoder bias, center-of-mass position, link masses and inertias, motor gains, and command delay. Plate dimensions remain fixed during training, while plate mass is randomized.

V. Experiments

We compare our method with existing humanoid recovery methods in simulation, analyze the geometric reward and two execution costs, and evaluate the deployed policy among physical obstacles.

A. Experimental Setup

Conditions. We evaluate three simulated conditions. *Flat* is unobstructed ground. *Guided* uses a $0.90 \times 0.64 \times 0.07$ m, 6 kg plate that can be lifted but cannot translate sideways or rotate, making lateral movement important for creating room to rise. *Free-R* uses a free plate whose length and width are each scaled by a factor in $[1, 1.75]$, with mass in $[6, 18]$ kg, horizontal offset within ± 0.15 m and yaw within $\pm 45^\circ$; training uses only the nominal size, so larger plates are unseen. Every trial starts from a low ($\mathcal{L}$) state, with supine, prone, left-side and right-side starts equally likely and 25% of states drawn from the procedural fallen-posture families. Each obstacle condition contains 512 trials and Flat contains 1024. All methods share the same initial states and Free-R draws. Evaluation disables external pushes and observation noise.

Metrics. A trial succeeds (SR) if the robot reaches standing within 20 s and then remains standing for 3 s. Standing requires a head height of at least 1.15 m, base uprightness of at least 0.90, and a vertical ground reaction above 20 N on each foot. We also measure execution quality over the full 23 s window, including failed trials. *Slip* is the travel of contact feet per second while the robot is near upright (head above 0.78 m, uprightness above 0.70). HLS_{95} is the 95th percentile, across trials, of the time the most loaded joint spends above 70% of its torque limit while moving slower than 0.3 rad/s.

Policies. For each training configuration, we report the checkpoint with the highest combined success under the guided plate and a fixed-size free plate among its training runs.

TABLE III

Comparison with humanoid recovery methods (SR, % ↑).

Method	Flat	Guided	Free-R
HoST [2]	98.7±0.7	59.8±4.2	80.9±3.4
HumanUP [1]	97.6±0.9	34.4±4.1	74.4±3.8
AMP [31]	68.1±2.9	2.1±1.3	4.9±1.9
Ours	**100.0***	**88.5±2.8**	**98.6±1.0**

Baselines use their original training configurations and are evaluated without retraining. All methods share the same initial states and Free-R draws. ± gives the 95% confidence half-width (normal approximation) over 1024 flat and 512 plate trials. *1024/1024 trials; Wilson 95% interval [99.6, 100].

B. Comparison with Recovery Methods

We compare with HoST [2] and HumanUP [1], two learning-based get-up methods for humanoids, and with an open-source G1 controller that learns locomotion and fall recovery in a single policy using an adversarial motion prior [17], [31]. All three use their original training configurations, which contain no overhead obstacles, and are evaluated in our conditions without retraining. This comparison tests whether recovery skills learned in open space carry over to overhead constraints.

HoST and HumanUP already stand from these initial states on open ground, so their lower success under the plates indicates difficulty adapting these recovery skills to overhead constraints (Table III). AMP fails on a substantial share of the same flat trials, which suggests that many of those states lie outside the distribution it was trained on. The gap is largest in Guided: the plate cannot be pushed aside, and a reward for raising the torso likely drives the body into it. In Free-R an upward motion can shove the plate away, so an open-space get-up often still succeeds. Our policy is trained to move out while staying low, and it leads in both conditions.

C. Ablation of the Training Objective

Table IV removes one reward component at a time and keeps the other terms unchanged. Each row reports that configuration's selected checkpoint.

In Guided, the plate can only be lifted, so the robot usually has to move out from under it before it can stand. Without the geometric reward, the escape reward retains only escape completion and the excess-plate-force cost. Escape completion pays only after sustained clearance, which the robot rarely discovers when the plate cannot be displaced, so Guided success falls sharply (Table IV). In Free-R the plate can also be pushed, tilted or lifted. The geometric reward helps here as well, because covered depth falls whether the body leaves the plate or the plate is pushed off the body; it rewards displacing the plate in a way that frees the

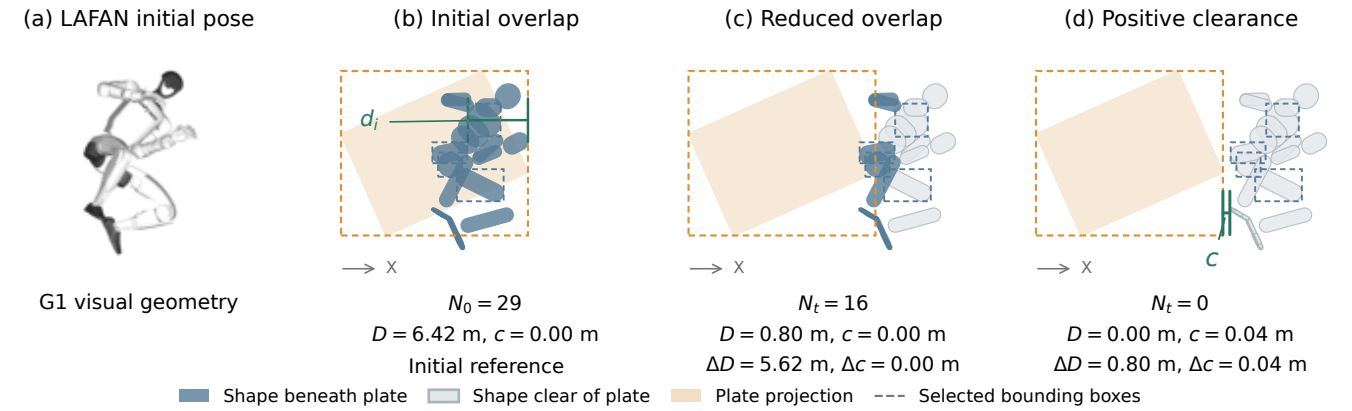


Fig. 3. Geometric measures for a supine LAFAN1 pose. (a) G1 geometry. (b–d) The same pose translated relative to a yawed, tilted plate, with selected world-axis-aligned bounds. N counts the collision shapes beneath the plate (blue), d_i is the depth of one shape, and D sums the depths over all 29 shapes. N and D decrease before a positive gap c appears. The configurations are constructed for illustration and are not taken from a rollout.

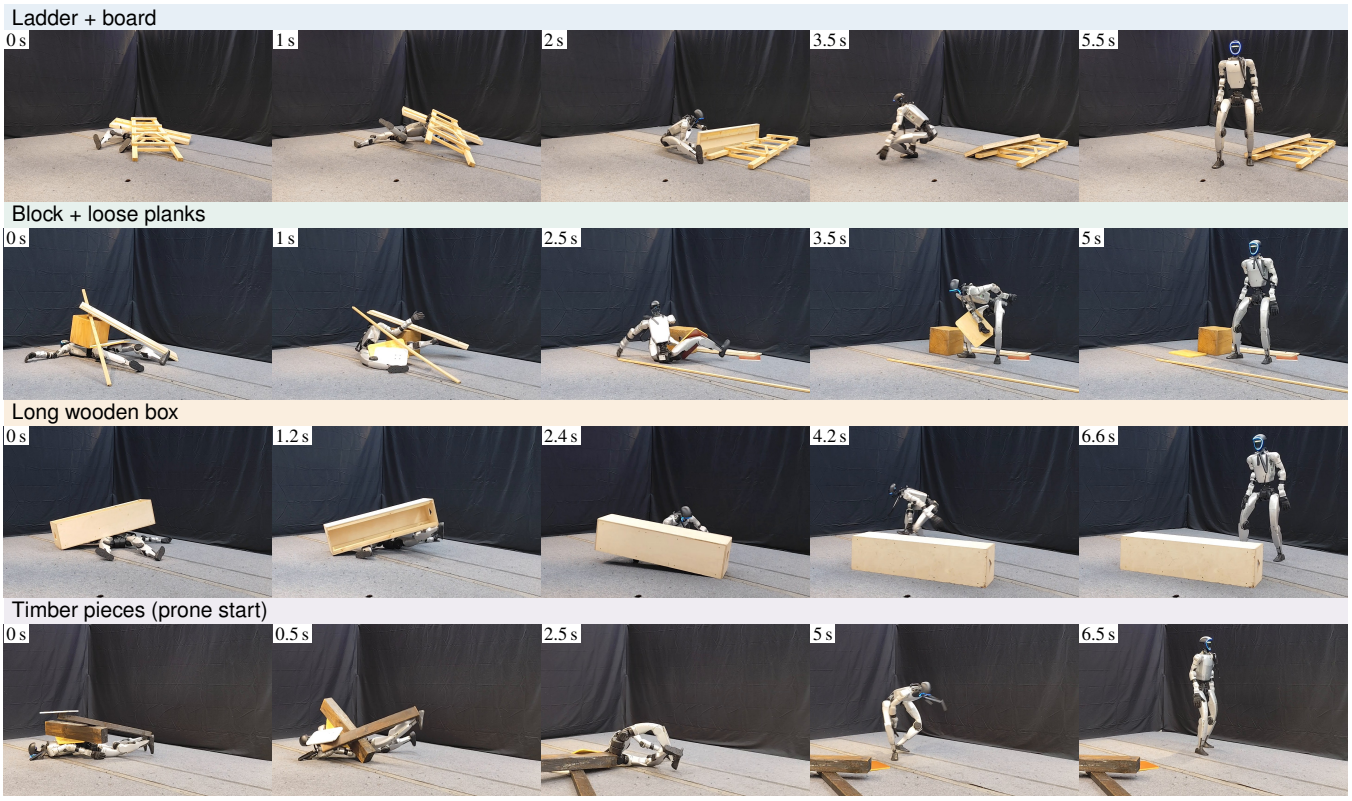


Fig. 4. Hardware recovery among obstacle arrangements absent from training: a ladder and board, a block with loose planks, a long wooden box, and timber pieces. The first three rows start supine and the bottom row starts prone. Each row shows frames from one trial; timestamps are relative to the first frame shown.

body. The foot-motion and joint-stall costs do a different job. Policies trained without either cost retain substantial Guided success but have higher measured foot slip and joint-stall duration. Slip and HLS_{95} measure the behaviors these costs target, so this increase is expected; the indirect effects are more informative. Removing L also doubles foot slip, and removing Q lowers Free-R success, so the two costs act together on the end of the rise rather than on separate metrics. To test sensitivity to model error, we increase upper-body mass and inertia by 30% without retraining; Guided success decreases from 88.5% to 85.5%.

D. Hardware Experiments

We deploy a policy trained with the complete objective on the Unitree G1 and test it in eight scenes (Table V): flat ground as a control, four single obstacles, and three arrangements of multiple objects. Apart from the single board, these obstacles differ from the training plates in shape, mass and constraint. In each attempt, the robot is placed beneath the obstacle with a randomly chosen heading, and attempts within a scene are run consecutively. The reported hardware SR counts use a 20 s deadline to reach standing and require at least 3 s of uninterrupted stable standing, with

TABLE IV

ABLATION OF THE TRAINING OBJECTIVE.

Configuration	SR (%) ↑ Guided	SR (%) ↑ Free-R	Slip ↓ (mm/s)	HLS_{95} ↓ (s)
Ours	**88.5±2.8**	**98.6±1.0**	**5.8±0.3**	**16.0±1.0**
w/o geometric reward	18.9±3.4	83.0±3.3	8.6±0.5	18.7±1.7
w/o foot-motion cost Q	83.2±3.2	93.8±2.1	19.2±0.5	19.3±0.5
w/o joint-stall cost L	84.6±3.1	97.3±1.4	12.6±0.3	21.3±0.3

Each configuration uses its checkpoint with the highest combined success under the guided plate and a fixed-size free plate. Slip and HLS_{95} pool both conditions. ± gives the 95% confidence half-width over evaluation trials (normal approximation for SR, bootstrap for Slip and HLS_{95}).

TABLE V

HARDWARE RECOVERY SUCCESS ACROSS OBSTACLE ARRANGEMENTS.

Scene	Successes/N	SR (%) ↑	95% CI
Flat ground (control)	20/20	100.0	[83.9, 100.0]
Single obstacle			
Single movable board	10/13	76.9	[49.7, 91.8]
Under table	27/35	77.1	[61.0, 87.9]
Rigid equipment case	12/16	75.0	[50.5, 89.8]
Long wooden box	12/14	85.7	[60.1, 96.0]
Multiple objects			
Board + ladder	7/29	24.1	[12.2, 42.1]
Block + loose planks	8/10	80.0	[49.0, 94.3]
Timber arrangement	19/33	57.6	[40.8, 72.8]
All obstacle scenes	95/150	63.3	[55.4, 70.6]

Attempt counts differ by scene. The bottom row excludes the flat-ground control. Intervals are Wilson 95% confidence intervals.

observation up to 23 s. The trials in Figs. 1 and 4 use the same policy.

The policy stands in every flat-ground attempt and in 95 of 150 obstacle attempts (Table V). Obstacle mobility appears to influence the observed recovery behavior. A board, a case or a long box can be pushed or tipped aside, and the robot then rises as it does under a free plate, though less reliably than in simulation. The table cannot be lifted or slid, and the illustrated recovery proceeds by rolling out past an edge (Fig. 1); unlike the guided plate, it cannot even be raised. Arrangements of several objects vary. Where the pieces leave an exit, as with loose planks on a block, the policy often gets out, although this scene has only ten attempts. Where a limb can be caught and held, as with a ladder under a board, it usually fails.

E. Failure Analysis

We observed board–ladder failures involving mechanical entanglement. As the robot pushes up beneath the board, an arm can pass between the ladder rungs. During the subsequent rise, a rung hooks the forearm, and the robot either lifts the ladder with it or stalls with the shoulder and elbow restrained. The timber arrangement produces similar trapping when a limb wedges between pieces. Beneath the table, the robot can catch on a corner and remain stuck short of the edge. Single-plate training does not cover trapping by ladder rungs or gaps between multiple objects. The deployed observations provide no explicit representation of this contact geometry.

VI. DISCUSSION AND CONCLUSION

We presented a learning approach for humanoid recovery beneath overhead obstacles. A geometric reward encourages the robot to create room to rise, while a motion prior and posture-dependent costs shape the motion. A single proprioceptive policy transfers to physical scenes that differ from the single-plate training distribution, including furniture and multi-object arrangements.

Limitations. The current policy does not reliably resolve limb entanglement in the tested scenes (Sec. V-E). Training uses a single plate, and proprioceptive observations provide no explicit representation of surrounding contact geometry. Covered depth and the clearance gap approximate tilted geometry with axis-aligned bounds. Training with multiple interacting objects and adding exteroceptive or tactile feedback are natural next steps.

REFERENCES

[1] X. He, R. Dong, Z. Chen, and S. Gupta, "Learning getting-up policies for real-world humanoid robots," in *Proceedings of Robotics: Science and Systems*, 2025.

[2] T. Huang, J. Ren, H. Wang, Z. Wang, Q. Ben, M. Wen, X. Chen, J. Li, and J. Pang, "Learning humanoid standing-up control across diverse postures," in *Proceedings of Robotics: Science and Systems*, 2025.

[3] Z. Xu, Y. Li, K.-Y. Lin, and S. X. Yu, "Unified humanoid fall-safety policy from a few demonstrations," in *IEEE International Conference on Robotics and Automation*, 2026.

[4] O. Azulay, Z. Xu, A. Scheffer, and S. X. Yu, "Unified perceptive fall safety for humanoids," *arXiv preprint arXiv:2602.16511*, 2026.

[5] X. Xu, Z. Chen, Y. Zhao, X. Zhang, R. Song, and W. Zhang, "UniReLo: Learning a unified humanoid policy from fall recovery to locomotion across diverse terrains," *arXiv preprint arXiv:2606.08922*, 2026.

[6] J. Morimoto and K. Doya, "Acquisition of stand-up behavior by a real robot using hierarchical reinforcement learning," *Robotics and Autonomous Systems*, vol. 36, no. 1, pp. 37–51, 2001.

[7] J. Hwangbo, J. Lee, A. Dosovitskiy, D. Bellicoso, V. Tsounis, V. Koltun, and M. Hutter, "Learning agile and dynamic motor skills for legged robots," *Science Robotics*, vol. 4, no. 26, p. eaau5872, 2019.

[8] T. Haarnoja, B. Moran, G. Lever, S. H. Huang, D. Tirumala, J. Humplik, M. Wulfmeier, S. Tunyasuvunakool, N. Y. Siegel, R. Hafner, M. Bloesch, K. Hartikainen, A. Byravan, L. Hasenclever, Y. Tassa, F. Sadeghi, N. Batchelor, F. Casarini, S. Saliceti, C. Game, N. Sreendra, K. Patel, M. Gwira, A. Huber, N. Hurley, F. Nori, R. Hadsell, and N. Heess, "Learning agile soccer skills for a bipedal robot with deep reinforcement learning," *Science Robotics*, vol. 9, no. 89, p. eadi8022, 2024.

[9] S. Ha and C. K. Liu, "Multiple contact planning for minimizing damage of humanoid falls," in *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2015.

[10] S. Wang and K. Hauser, "Unified multi-contact fall mitigation planning for humanoids via contact transition tree optimization," in *IEEE-RAS International Conference on Humanoid Robots*, 2018, pp. 33–39.

[11] S. Noda, S. Nozawa, Y. Kakiuchi, K. Okada, and M. Inaba, "Contact-motion planning integrating multiple contact transition strategies and its application to slide contact motion for humanoid robot," *Journal of the Robotics Society of Japan*, vol. 35, no. 5, pp. 393–402, 2017.

[12] Y. Ma, F. Farshidian, and M. Hutter, "Learning arm-assisted fall damage reduction and recovery for legged mobile manipulators," in *IEEE International Conference on Robotics and Automation*, 2023, pp. 12 149–12 155.

[13] Z. Liu, "RecoverFormer: End-to-end contact-aware recovery for humanoid robots," *arXiv preprint arXiv:2604.22911*, 2026.

[14] C. Gonzalez and L. Sentis, "Whole-body planning for humanoids navigating confined spaces via self-collision avoidance references," *arXiv preprint arXiv:2608.10220*, 2026.

[15] X. B. Peng, P. Abbeel, S. Levine, and M. van de Panne, "DeepMimic: Example-guided deep reinforcement learning of physics-based character skills," *ACM Transactions on Graphics*, vol. 37, no. 4, pp. 143:1–143:14, 2018.

[16] J. Ho and S. Ermon, "Generative adversarial imitation learning," in *Advances in Neural Information Processing Systems*, vol. 29, 2016.

[17] X. B. Peng, Z. Ma, P. Abbeel, S. Levine, and A. Kanazawa, "AMP: Adversarial motion priors for stylized physics-based character control," *ACM Transactions on Graphics*, vol. 40, no. 4, 2021.

[18] A. Escontrela, X. B. Peng, W. Yu, T. Zhang, A. Iscen, K. Goldberg, and P. Abbeel, "Adversarial motion priors make good substitutes for complex reward functions," in *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2022, pp. 25–32.

[19] X. B. Peng, Y. Guo, L. Halper, S. Levine, and S. Fidler, "ASE: Large-scale reusable adversarial skill embeddings for physically simulated characters," *ACM Transactions on Graphics*, vol. 41, no. 4, 2022.

[20] C. Tessler, Y. Kasten, Y. Guo, S. Mannor, G. Chechik, and X. B. Peng, "CALM: Conditional adversarial latent models for directable virtual characters," in *ACM SIGGRAPH Conference Proceedings*, 2023.

[21] Y. Lu, Y. Zhong, L. Zhao, W. Li, and P. Lu, "Unified walking, running, and recovery for humanoids via state-dependent adversarial motion priors," *arXiv preprint arXiv:2605.18611*, 2026.

[22] W. Wu, M. Huang, R. Liu, J. Nie, S. Guo, and C. Li, "StableMimic: Smooth human-like recovery for humanoid motion tracking—learning beyond the tracking distribution for structured post-fall behavior," *arXiv preprint arXiv:2608.02385*, 2026.

[23] Z. Tao, Z. Su, P. Liu, J. Sun, W. Que, J. Ma, J. Yu, J. Cao, P. Sun, H. Liang, G. Han, W. Zhao, Z. Xu, J. Tang, Q. Zhang, and Y. Guo, "Heracles: Bridging precise tracking and generative synthesis for general humanoid control," *arXiv preprint arXiv:2603.27756*, 2026.

[24] C. Chi, S. Feng, Y. Du, Z. Xu, E. Cousineau, B. Burchfiel, and S. Song, "Diffusion policy: Visuomotor policy learning via action diffusion," in *Proceedings of Robotics: Science and Systems*, 2023.

[25] Y. Mu, Z. Zhang, Y. Shi, D. Yang, M. Matsumoto, K. Imamura, G. Tevet, C. Guo, M. Taylor, C. Shu, P. Xi, and X. B. Peng, "SMP: Reusable score-matching motion priors for physics-based character control," *ACM Transactions on Graphics*, vol. 45, no. 4, pp. 84:1–84:21, 2026.

[26] X. B. Peng, M. Andrychowicz, W. Zaremba, and P. Abbeel, "Sim-to-real transfer of robotic control with dynamics randomization," in *IEEE International Conference on Robotics and Automation*, 2018.

[27] Z. Chen, X. He, Y.-J. Wang, Q. Liao, Y. Ze, Z. Li, S. S. Sastry, J. Wu, K. Sreenath, S. Gupta, and X. B. Peng, "Learning smooth humanoid locomotion through Lipschitz-constrained policies," in *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2025.

[28] E. Todorov, T. Erez, and Y. Tassa, "MuJoCo: A physics engine for model-based control," in *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2012, pp. 5026–5033.

[29] F. G. Harvey, M. Yurick, D. Nowrouzezahrai, and C. Pal, "Robust motion in-betweening," *ACM Transactions on Graphics*, vol. 39, no. 4, 2020. [Online]. Available: https://github.com/ubisoft/ubisoft-laforge-animation-dataset

[30] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," *arXiv preprint arXiv:1707.06347*, 2017.

[31] ccrpRepo, "AMP_mjlab: G1 AMP motion control built on mjlab and rsl_rl," https://github.com/ccrpRepo/AMP_mjlab, accessed Oct. 20, 2026.